

PROJECT DOCUMENTATION

Network Request Management System

Functional & System Documentation

Field	Details
Project Title	Network Request Management System
Project Type	Individual Project
Platform	ServiceNow
Documentation	FSD / Project Documentation

1. Introduction

This document describes the Network Request Management and Fulfillment System implemented on the ServiceNow platform. It follows the supplied project documentation format, which organizes the documentation into introduction, project overview, architecture, setup, structure, running the application, API documentation, authentication, user interface, testing, screenshots/demo, known issues and future enhancements.

Project Title: Network Request Management and Fulfillment System

Platform: ServiceNow

Objective: To provide a standardized, role-controlled and trackable lifecycle for network-related service requests from submission and validation through approval, fulfillment and completion.

2. Project Overview

The project provides a Network Request catalog item through the Service Catalog/Service Portal. It captures request details such as request type, access level, device and business justification. Client-side logic dynamically displays relevant fields based on the selected request type and validates request information.

The submitted request is routed for approval based on network access type and priority. Approvers receive automated email notifications. Once approved, the system creates network fulfillment tasks and assigns them to the appropriate Network Team. Network engineers process the task, update work notes and complete the required configuration or access change. Request status and requester notifications are updated throughout the lifecycle.

2.1 Key Features

- Network Request catalog item through Service Portal / Service Catalog.
- Request type, access level, device and business justification capture.
- Dynamic field display based on selected network request type.
- Client-side validation of request information.
- Groups, roles and access controls for Requesters, Approvers and Network Team.
- Approval routing based on network access type and priority.
- Automated approval and requester email notifications.
- Automatic network fulfillment task creation and assignment.
- Engineer work notes and task completion.
- Request lifecycle tracking: New, In Progress, Awaiting Approval and Completed.
- Lists, reports, dashboards and SLA compliance monitoring.

3. Architecture

The application is implemented natively on ServiceNow. ServiceNow provides the user interface, request records, catalog configuration, client-side logic, role-based access, workflow/Flow Designer automation, approvals, notifications, fulfillment tasks and reporting. A separate React frontend, Node.js/Express backend or MongoDB database is not part of the current project scope.

3.1 Architecture and Data Flow

The architecture follows the complete lifecycle: Requester → Service Portal/Service Catalog → Network Request catalog item → dynamic field logic and validation → approval routing → approval decision → fulfillment task creation → Network Team assignment → Network Engineer processing → request status and notifications → administrator/developer monitoring.

Requester	Service Catalog	Client Logic	ServiceNow Request
Submit	Network Request	Dynamic Fields + Validation	Request Record
Approval Engine	Approver	Email Notification	Approval Record
Fulfillment Task	Network Team	Network Engineer	Work Notes / Completion
Request Status	Requester Notification	Reports & Dashboards	SLA Monitoring

Figure 1: ServiceNow Network Request architecture and data flow

3.2 Architecture Components

Component	Purpose
Service Portal / Service Catalog	Entry point for Requesters to submit the Network Request.
Network Request Catalog Item	Captures request type, access level, device, business justification and related variables.
Client-side Logic	Dynamically displays applicable fields and validates input based on request type.
Roles and Access Controls	Control access for Requesters, Approvers, Network Team members, Administrators and Developers.
Flow Designer / Workflow	Automates approval routing, notifications and fulfillment task creation.
Approval Records	Store the approval stage and decision.
Fulfillment Tasks	Provide actionable work items to the Network Team and Network Engineers.
Reports / Dashboards / SLA	Support monitoring of requests, approvals, task performance and SLA compliance.

4. Setup Instructions

For this ServiceNow project, setup is performed in a ServiceNow instance rather than through local npm installation. The supplied documentation format asks for prerequisites and installation/environment setup; those concepts are adapted here to the ServiceNow platform.

4.1 Prerequisites

- Access to a ServiceNow instance/PDI with appropriate administrator/developer permissions.
- Service Catalog / Service Portal functionality available.
- Permission to configure catalog items, variables, client-side logic, groups, roles, approvals, workflows/flows, notifications and reports.
- Supported web browser for accessing the ServiceNow instance.

4.2 Configuration Sequence

1. Configure the Network Request catalog item and its variables.
2. Configure dynamic field behavior and client-side validation.
3. Create Network Team, Approvers and Requesters groups and configure roles/access.
4. Configure approval routing based on access type and priority.
5. Configure approval and requester notifications.
6. Configure automatic fulfillment task creation and assignment.
7. Configure task/request status progression.
8. Configure reports, dashboards and SLA monitoring.
9. Test the complete lifecycle from submission to completion.

5. Application / Configuration Structure

Because the project is implemented in ServiceNow, the equivalent of a MERN client/server folder structure is the platform configuration structure below.

Configuration Area	Project Elements
Service Catalog	Network Request catalog item, variables and request capture.
Client-side Configuration	Client Scripts / dynamic field behavior / validation.
Access Control	Groups, roles and access controls.
Automation	Flow Designer / workflow for approvals, notifications and fulfillment.
Server-side Logic	Business Rules and platform logic used in the implementation.
Notifications	Approval notifications and requester lifecycle notifications.
Task Management	Network fulfillment tasks, assignment, work notes and completion.
Reporting	Lists, reports, dashboards and SLA monitoring.

6. Running the Application

The application is used through the ServiceNow Service Portal or Service Catalog. The normal execution flow is:

1. Requester opens the Network Request catalog item.

2. Requester selects the request type and enters required details.
3. Relevant fields are displayed dynamically and input is validated.
4. Requester submits the request.
5. ServiceNow routes the request for approval according to access type and priority.
6. Approver receives the notification and approves or rejects the request.
7. After approval, a network fulfillment task is automatically created.
8. The task is assigned to the appropriate Network Team.
9. Network Engineer processes the task, updates work notes and performs the required change.
10. Request/task statuses and requester notifications are updated through the lifecycle.
11. Administrators and Developers monitor the process using lists, reports, dashboards and SLA views.

7. API Documentation

The supplied format requests documentation of backend endpoints, methods, parameters and responses. The current project flow does not specify a custom REST API or separate backend service. The solution is implemented through ServiceNow catalog records, workflow/Flow Designer, approvals, notifications and task records.

Interface	Current Implementation
Request submission	Service Catalog / Service Portal; no custom REST endpoint specified.
Approval processing	ServiceNow approval records and configured workflow/Flow Designer.
Fulfillment	ServiceNow task records and assignment.
Notifications	ServiceNow email notification mechanism.
Reporting	ServiceNow lists, reports and dashboards.

If an external integration or custom API is added in a future version, endpoint, method, parameters, authentication and example responses should be documented here.

8. Authentication and Authorization

The supplied format includes authentication and authorization as a separate section. In this project, authentication and authorization are handled by the ServiceNow platform and its user, group, role and access-control model.

- Requesters receive the permissions needed to submit and view their requests.
- Approvers receive the permissions required to act on approval records assigned to them.
- Network Team members receive access required to work on assigned fulfillment tasks.
- Administrators and Developers receive the permissions required for configuration and monitoring according to their assigned roles.
- Access controls restrict records or operations to authorized users.

No separate JWT or custom session-management implementation is defined in the current project scope.

9. User Interface

The main user interface is the ServiceNow Service Portal / Service Catalog. The Network Request catalog item captures the required information and uses dynamic behavior to display relevant fields based on request type.

9.1 Requester Interface

- Network Request catalog item

- Request type selection
- Access level and device information
- Business justification
- Context-dependent fields
- Submission and lifecycle visibility

9.2 Approver Interface

- Approval record with request information
- Approve / Reject action
- Automated approval notification
- Access type and priority context

9.3 Network Engineer Interface

- Assigned fulfillment task
- Task details and ownership
- Work notes
- Task update and completion

9.4 Administrator / Developer Interface

- Request and task lists
- Approval tracking
- Reports and dashboards
- SLA compliance monitoring
- Configuration and access management

10. Testing

Testing focuses on request creation, client-side behavior, validation, approval routing, notifications, task generation, assignment, status progression and monitoring.

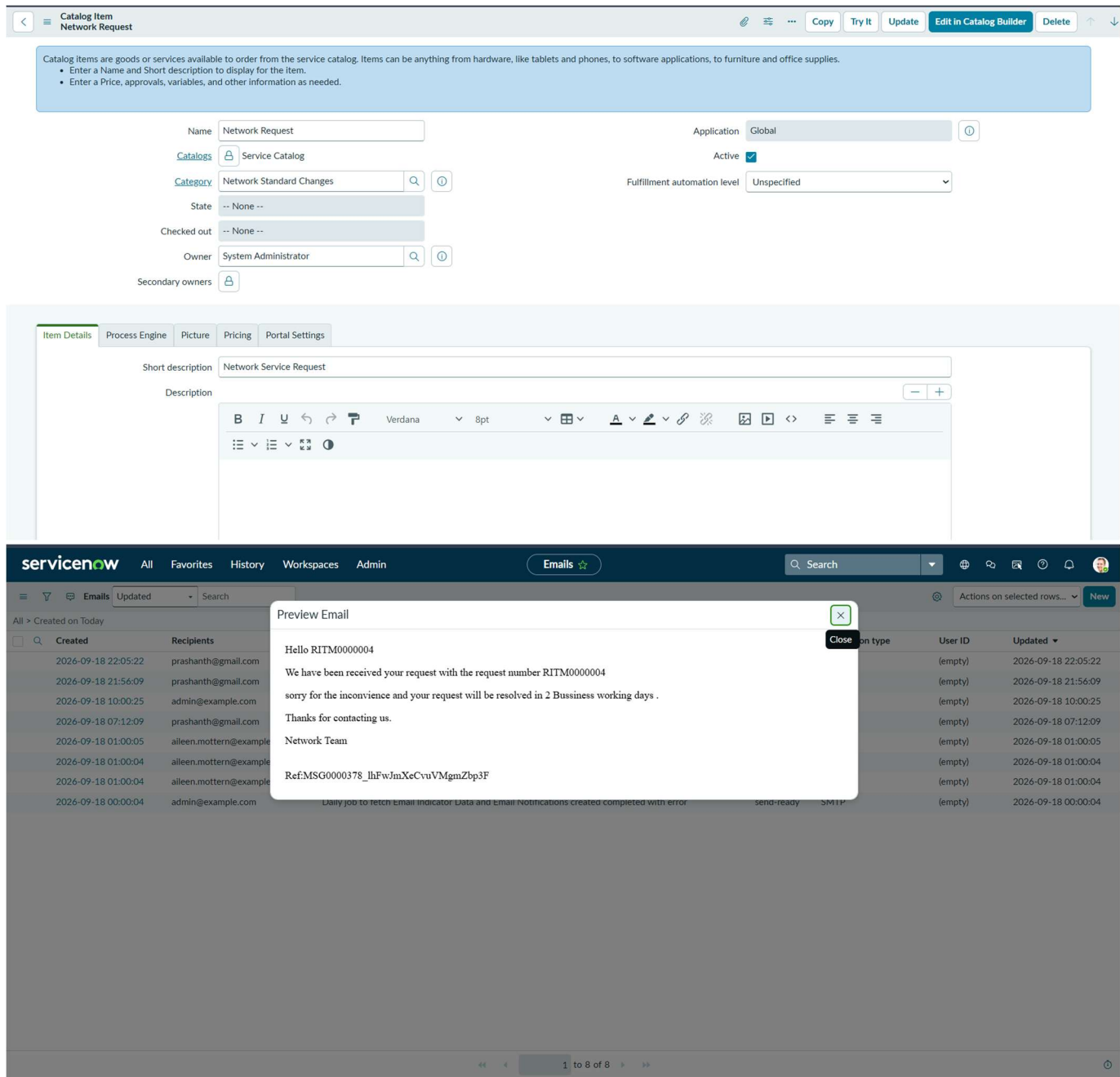
Test Case	Expected Result
Valid request submission	Request is created with required details.
Request type change	Applicable fields are displayed dynamically.
Invalid/incomplete data	Configured validation prevents incorrect submission.
Approval routing	Correct approver receives the request.
Approval notification	Approver receives the configured notification.
Request rejection	Request follows the configured rejection path and requester is notified.
Request approval	Fulfillment task is automatically created.
Task assignment	Task is assigned to the appropriate Network Team.
Work notes	Engineer updates are stored on the task.
Task completion	Task/request move to the configured completed state.
Lifecycle notification	Requester receives notifications at configured stages.

Reports/SLA

Records are available for monitoring and SLA tracking.

11. Screenshots / Demo

The supplied documentation format includes a screenshots/demo section. For the final submission, screenshots from the actual ServiceNow instance should be inserted at the locations below.



Database Tables
Created 2026-09-18 22:05:22

Update
Delete

Database Number

RITM0000004

Assignment to

ITIL User

Mobile Number

Total Amount

Mode of Payment

Type of Connection

Customer Address

Approval Status

Approved

Requested For

Update

Delete

Approval Request

State

Search

Actions on selected rows...
New

Approvals

State	Approver	Comments	Approval for	Created
Approved	System Administrator		(empty)	2026-09-18 22:05:22

1 to 1 of 1

Workflow Studio

Network Request

Network Request

Flow execution - None

EXECUTION DETAILS

Network Request

Test Run - Completed

Open flow

Open context record

Show Action Details

Run as: System Administrator

Open flow logs

Completed

2026-09-18 22:05:22

349ms

FLOW STATISTICS

TRIGGER

Catalog Item Requested

ACTIONS

1	Get Catalog Variables from Network Request	Core Action	Completed	2026-09-18 22:05:22	27ms
2	Create Record	Core Action	Completed	2026-09-18 22:05:22	7ms
3	Send Email		Completed	2026-09-18 22:05:22	45ms
4	Ask For Approval	Core Action	Completed	2026-09-18 22:05:22	255ms
5	If if Request is Approved	Flow Logic	Evaluated - True	2026-09-18 22:06:39	9ms
6	Update Record	Core Action	Completed	2026-09-18 22:06:39	9ms
7	Else	Flow Logic	Not Run		
8	Update Record	Core Action	Not Run		

12. Known Issues

The provided project flow does not identify any unresolved bugs or production defects. Therefore, no unverified issue is recorded as a known issue.

Before final submission, instance-specific details such as exact field names, role names, SLA definitions and final screenshots should be checked against the implemented ServiceNow instance.

13. Future Enhancements

- Add additional network request categories and standardized request templates.
- Introduce more granular approval matrices for departments, risk levels and access types.
- Add management-level SLA and performance dashboards.
- Integrate with external network-management or identity systems when required.
- Add reminders and escalations for pending approvals and overdue tasks.
- Add knowledge articles and contextual help to improve request quality.
- Reuse the workflow pattern for other IT service requests such as hardware, software and access provisioning.

14. Conclusion

The Network Request Management and Fulfillment System provides an end-to-end ServiceNow workflow for network-related requests. It connects standardized request capture, dynamic field behavior, validation, role-based access, approval routing, notifications, fulfillment task creation, engineer processing, lifecycle status tracking and operational monitoring. The result is a structured process that gives Requesters visibility, Approvers controlled decision points, Network Engineers actionable tasks, and Administrators/Developers centralized monitoring.